

Adversarially Asymmetric Evidence for Calibrated Fake News Detection
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Abstract

A reader who receives a news story as a link, a forwarded message or a screenshot must decide whether to believe it. Systems built for this purpose score the language of an article: its tone, its headline, and whether it names its sources. Each signal is selected by the author, so a fabrication written in conventional newsroom register attains the same score as a genuine report; on such a design, invented articles averaged 7.3 out of 10 and four of six were presented to readers as credible. This work treats detection as an adversarial problem and adopts the principle that follows from it: an observation under the author's control may count against an article but never in its favour. Support is therefore admitted only from outside the author's control, namely corroboration by independent newsrooms, counted in distinct sources rather than retrieved articles so that common ownership and syndicated copy collapse, together with verification of the article's premises against a reference work where a news index cannot adjudicate. Each observation contributes a likelihood ratio estimated from labelled articles rather than a weight assigned by hand, and these combine into a calibrated probability. The system operates across languages, declines input too unspecific to verify and reports what is absent, returns a definite verdict rather than a score, and assesses writing quality separately, since a well-written account may still be false. On held-out data, accuracy increased from 0.888 to 0.954 and precision among flagged articles from 0.848 to 0.975.

Keywords — fake news detection, adversarial evidence, likelihood ratio, probability calibration, source independence, claim verification

Contributions

1. An adversarial admissibility rule under which observations controlled by the author of a text are admitted only in the incriminating direction, so that presentation cannot establish credibility.
2. Corroboration measured in independent sources rather than in retrieved articles, with common ownership, agency syndication and near-duplicate reprints collapsed before any count is taken.
3. Verification of an article's premises against a reference work, conducted separately from verification of its events against news coverage, addressing claims that a news index cannot adjudicate.
4. Likelihood ratios estimated from labelled data, each traceable to the counts that produced it, in place of weights assigned by hand.
5. An input admissibility gate that declines content containing no verifiable claim and states the information required to make it verifiable.

6. Separation of veracity from presentation quality, reported as distinct outputs rather than combined into a single score.

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